INKFLOW: Iterative Narrative Knowledge for Manga Generation

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Abstract

This project proposes an automated system that converts a textual story (PDF) into a sequence of manga-style panels. The pipeline uses a large language model (LLM) to split the story into short visual units and to generate structured prompts for a diffusion-based text-to-image model. To ensure that the generated panels follow the story and maintain a consistent manga style, a vision-language model (VLM) is used as a judge. The VLM evaluates story-image alignment and triggers prompt refinement and image regeneration when quality requirements are not met. The goal of this project is to build a reproducible framework for controlled text-to-image generation with improved narrative consistency across panels.

Keywords

manga generation, diffusion models, prompt control, vision-language models, cross-modal alignment

1 Project Topic and Number

Text-to-Image Generation and Prompt Control (Topic #9), and *Cross-Modal Consistency and Alignment Analysis (Topic #12)*.

2 Problem Statement

Text-to-image diffusion models are capable of generating visually appealing images, but they often struggle with following longer narratives, maintaining consistent characters across multiple images, and preserving a coherent visual style. These issues become more pronounced when generating manga panels from a story, where even a small mistake in character appearance, actions, or props can break the narrative flow. Existing approaches typically rely on manual prompt tuning and repeated human inspection, which is time-consuming and difficult to reproduce. This project addresses the problem of enabling automated, prompt-controlled manga generation with built-in verification to reduce mismatches between story text and generated panels.

3 Proposed Technique

The proposed system converts a story PDF into manga panels using an automated LLM-diffusion-VLM pipeline:

- **PDF ingestion and beat segmentation:** Text is extracted from a story PDF (using OCR if necessary), and an LLM divides the story into short narrative beats of one to three sentences. Each beat is represented in a structured form that includes setting, characters, actions, emotions, props, and dialogue.
- **Series bible / memory:** A persistent series bible is maintained in JSON format to store canonical information about characters, such as appearance and outfits, as well as recurring props and locations. This information is reused to maintain visual consistency across panels.
- **Prompt-controlled diffusion generation:** For each beat, an LLM generates a structured prompt tailored for manga-style image generation, including style cues, camera framing, and composition details. Optional control signals, such as pose or layout constraints, may be used to improve alignment with the narrative beat.
- **Post-render dialogue:** Speech bubbles and dialogue text are rendered programmatically after image generation to ensure clear, readable text and consistent typography, avoiding common text artifacts produced by diffusion models.
- **VLM-based judging and refinement loop:** A vision-language model evaluates each generated panel with respect to the narrative beat the series bible, and previously generated panels. The evaluation considers story correctness, character consistency, manga style, and text readability. If a panel does not meet predefined thresholds, the prompt is refined and the panel is regenerated for a limited number of attempts.

To support reproducibility, the system logs prompts, random seeds, control parameters, evaluation scores, and refinement history for each panel.

4 Expected Outcomes

The expected outcomes of this project include:

- **End-to-end prototype:** A working pipeline that takes a short public-domain story PDF as input and produces a sequence of manga panels and compiled pages.
- **Quantitative evaluation:** Automatically computed metrics and judge scores for each panel, including story-image alignment scores from the VLM, OCR-based text accuracy for dialogue, and basic checks for character consistency.
- **Ablation study:** An analysis of how different pipeline components affect performance, including the use of a series bible, the VLM-based refinement loop, pose or layout controls, and different choices of LLM, diffusion model, and VLM configured through a unified configuration file.
- **Reproducible artifacts:** Structured outputs such as beat representations and series bible files that enable systematic experimentation and comparison of prompt-controlled text-to-image generation pipelines.

5 Scope and Feasibility

The initial implementation will focus on a single short public-domain children's story and generate approximately 8-20 manga panels. The system can be extended to support more complex page layouts and stronger continuity constraints if time permits. The project is feasible within a semester because each component of the pipeline can be developed and tested independently while integrating into a unified workflow.